SHL Conversational Assessment Recommender — Approach Summary

1. Problem Statement

The goal of this project was to build a conversational SHL assessment recommendation system that helps recruiters and hiring managers discover relevant SHL assessments using natural language queries instead of traditional keyword-based filtering. The system accepts conversational hiring requirements and returns relevant SHL assessments using vector-based retrieval techniques.

2. System Design

The project was implemented using a Retrieval-Augmented Generation (RAG)-style architecture. Main components include FastAPI backend, Sentence Transformer embeddings, FAISS vector similarity search, and catalog-based recommendation retrieval. The API exposes GET /health and POST /chat endpoints with Swagger documentation support.

3. Retrieval Setup

SHL assessment catalog data was stored in catalog.json. Assessment descriptions were converted into embeddings using sentence-transformers/all-MiniLM-L6-v2. Embeddings were indexed using FAISS. User queries were converted into embeddings and similarity search was used to retrieve the most relevant assessments. Duplicate recommendations were filtered before returning responses.

4. Prompt and Conversation Design

The API accepts conversation history in structured JSON format. Basic clarification handling was implemented for vague queries by prompting users to provide more role or skill details. The response structure includes conversational reply, recommendations, and conversation status flag.

5. Evaluation Method

The system was evaluated using manual testing and retrieval validation techniques. Evaluation included recommendation relevance testing, duplicate filtering validation, clarification behavior testing, groundedness validation, API response validation, and retrieval quality testing.

6. Challenges and Fixes

Several issues were encountered during development including empty catalog data, FAISS index path mismatches, duplicate recommendations, dependency conflicts, and deployment configuration issues. These were resolved through safer catalog handling, duplicate filtering logic, corrected FAISS indexing workflow, and environment upgrades.

7. Improvements Made

Improvements implemented during development included duplicate filtering, improved retrieval stability, improved FastAPI schema handling, clarification support, better embedding text quality, structured API responses, and improved recommendation consistency.

8. Future Improvements

Future enhancements include real-time SHL catalog scraping, better conversational memory, assessment comparison support, hybrid retrieval methods, reranking techniques, and optional LLM integration for grounded conversational responses.

9. Conclusion

The final system successfully implemented a conversational SHL assessment recommender using FastAPI, FAISS, and sentence-transformer embeddings. The project demonstrates semantic retrieval, conversational API handling, and grounded recommendation generation using catalog-based vector search.